



PGA TOURNAMENT OUTCOME PREDICTION USING PERFORMANCE METRICS

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1. INTRODUCTION

The goal of this project is to develop a predictive model that assigns a probability to a player's likelihood of finishing in specific positions within a PGA golf tournament. The model utilizes historical player performance data to make predictions about future tournament outcomes. Data for this analysis includes PGA Tour data from 2015 to 2022, sourced from Advanced Sports Analytics.

2. EXPLORATORY DATA ANALYSIS

DATA OVERVIEW

The dataset includes player-specific details such as player name, tournament, strokes, number of rounds, made cut, and final position. Additionally, it contains strokes gained metrics such as total strokes gained (SG), SG: tee-to-green, SG: off-the-tee, SG: approach, SG: around-the-green, and SG: putting. Each row of the dataset represents the performance of a player in a single tournament. The dataset comprises over 32,000 competitions by 500 players in the PGA Tour between the years of 2004-2022.

DATA EXPLORATION

Initially, the goal was to classify player finishes in specific categories, such as Top 10, Top 20, or Top 50, using binary classification. However, the data distribution was highly skewed, particularly toward players making the cut, making binary classification challenging. A stratified split strategy was considered but ultimately rejected in favor of a more effective approach: assigning probabilities to each player's likelihood of finishing in specific tournament positions.

One key observation from the data was the strong correlation between strokes gained values and a player's position in the tournament. Since a player's strokes gained essentially determines their final position, the model would rely heavily on strokes gained metrics from previous tournaments to predict future performance.

3. FEATURE ENGINEERING

ADJUSTED STROKES GAINED

A challenge with strokes gained data is its variability across different tournaments, making it difficult to compare player performance directly across events due to differing course conditions, competition levels, and other tournament-specific factors. To address this issue, an adjusted strokes gained metric was used to control for both player-specific skill levels and tournament-

specific effects, enabling a more accurate assessment of a player's performance relative to their peers across different events.

A fixed effects model was used to isolate these influences. In this model, each player is assigned a fixed effect that represents their average strokes gained across all tournaments, effectively capturing their baseline skill level. Additionally, each tournament is assigned a fixed effect, which controls for tournament-specific factors that could affect performance, such as course difficulty or weather conditions. These fixed effects collectively account for both individual player skill and the external tournament environment, creating a model that predicts what a player's strokes gained should be, based on these factors.

The adjusted strokes gained value is calculated as the residual (adjusted strokes gained total), which is the difference between a player's actual strokes gained and the predicted strokes gained from the model. This residual represents how much a player overperformed or underperformed compared to their expected performance, after controlling for their skill level and the conditions of the tournament.

By removing the biases caused by differences in player ability and tournament conditions, the adjusted strokes gained metric provides a clearer, more standardized view of a player's performance. It allows for a fair comparison of players across different tournaments, as it isolates the player's true performance from external influences. This adjustment forms the foundation for generating features in the predictive model, as it reflects each player's deviation from their expected performance, making it easier to identify trends, predict future outcomes, and compare performance across a range of events.

PREDICTIVE FEATURES

After calculating the adjusted strokes gained metric, several features were generated to improve the predictive power of the model:

1. Exponentially Weighted Moving Averages (EWMAs):

- EWMAs were calculated for each player's adjusted strokes gained over different time periods: 2 months, 1 year, and 2 years. These features capture short-, medium-, and long-term performance trends, with more recent tournaments played weighted more heavily.
- EWMAs are preferable to simple averages because they adapt to changes in a player's form by emphasizing recent performance. Simple averages, by contrast, can obscure important trends, as they treat all tournaments performances equally.

2. Rolling Averages and Rolling Standard Deviations:

- Rolling statistics for the adjusted strokes gained were calculated over 3-tournament and 6-tournament windows. Rolling means provide insight into short- and medium-term

performance trends, while rolling standard deviations assess the consistency of a player's performance.

- These features are particularly valuable for evaluating recent form and identifying patterns of variability in performance.

3. Last Event Score

- The last event score feature captures a player's adjusted strokes gained from their most recent tournament. By shifting the adjusted strokes gained metric by one event for each player, this feature gives insight into how well the player performed in their previous competition.
- This feature is useful in predicting future performance because recent form can have a significant impact on a player's upcoming results. A strong or weak showing in the last event provides context for predicting how a player might perform in their next tournament.

4. MODEL DEVELOPMENT & TESTING

TARGET VARIABLE

In this predictive model, the target variables represent different performance categories for players in a tournament. These categories are defined as binary classifications based on the player's finishing position. Specifically, the code creates five target variables: Top 10, Top 20, Top 40, Made Cut, and Missed Cut. Each target variable is assigned a value of 1 or 0, where 1 indicates that the player met the condition for that category, and 0 indicates that they did not.

- **Top 10:** If a player finishes in 10th place or better, they are assigned a 1; otherwise, they are assigned a 0.
- **Top 20:** If a player finishes in 20th place or better, they are assigned a 1; otherwise, they are assigned a 0.
- **Top 40:** If a player finishes in 40th place or better, they are assigned a 1; otherwise, they are assigned a 0.
- **Made Cut:** If a player finishes in 70th place or better, they are assigned a 1, signifying they made the cut.
- **Missed Cut:** If a player finishes worse than 70th place, they are assigned a 1 for missed cut, and 0 for making the cut.

For example, if a player finishes in 8th place in a tournament, they will be assigned a binary classification of 1 for the Top 10 target variable, since 8th place falls within the top 10.

Similarly, the player will also receive a 1 for the Top 20 and Top 40 target variables, as 8th place meets the conditions for both categories. The Made Cut variable will also be assigned a 1, as finishing in 8th place means the player successfully made the cut by finishing in the top 70. Conversely, the player will be assigned a 0 for the Missed Cut variable, since they did not miss the cut.

PERFORMANCE METRICS

Initially, a standard 70-30 train-test split was used to train and test the model. However, this method failed to reflect the practical application of the model, as it did not account for the temporal aspect of player performance. To address this, the model was revised to use the first 70% of competitions played by each player as training data, with the remaining competitions used for testing.

The performance of the model was evaluated using two metrics: **Brier Score** and **Log Loss**.

1. Brier Score:

Brier score measures the accuracy of probabilistic predictions by evaluating the mean squared difference between predicted probabilities and actual outcomes. It is particularly useful when the goal is to assess how well-calibrated the model's probabilities are, with lower values indicating better calibration. Calibration refers to how closely the predicted probabilities align with the actual observed outcomes. A well-calibrated model means that its predicted probabilities correspond to the true likelihood of events. For example, if a model predicts that a player has a 70% chance of making the cut, a well-calibrated model would show that players with a 70% predicted probability indeed make the cut around 70% of the time. In this way, calibration ensures that the model's probability estimates are reliable and realistic, rather than overconfident or underconfident in its predictions.

2. Log Loss:

Log loss evaluates the confidence of the model's predictions by heavily penalizing incorrect predictions that are made with high confidence. This makes log loss a more stringent measure, as it focuses on minimizing overconfident mistakes, which can be crucial in decision-making scenarios.

5. MODEL SELECTION & RESULTS

To establish a baseline, a simple model was developed that assigned the same probability to each player based on the frequency of occurrences of each position category within the dataset. This baseline model's performance was compared to more advanced models, such as Random Forest, Logistic Regression, SVM, Naive Bayes, and K-Nearest Neighbors.

Among these models, Logistic Regression consistently performed the best, with better overall performance compared to the baseline model. While the baseline model showed decent calibration (lower Brier Scores), it performed poorly in terms of Log Loss, indicating overconfidence in incorrect predictions.

Baseline Model Results:

Top 10: Brier Score: 0.0506 | Log Loss: 0.6386
Top 20: Brier Score: 0.0982 | Log Loss: 0.8779
Top 40: Brier Score: 0.1661 | Log Loss: 1.0753
Made Cut: Brier Score: 0.2384 | Log Loss: 1.0699
Missed Cut: Brier Score: 0.2384 | Log Loss: 1.0699

Logistic Regression Results:

Top 10: Brier Score: 0.0551 | Log Loss: 0.2723
Top 20: Brier Score: 0.0920 | Log Loss: 0.3649
Top 40: Brier Score: 0.1690 | Log Loss: 0.5536
Made Cut: Brier Score: 0.2444 | Log Loss: 0.7141
Missed Cut: Brier Score: 0.2444 | Log Loss: 0.7141

The Brier Score indicates that the baseline model is sometimes better calibrated, particularly in categories like "Top 10" and "Missed Cut." However, this calibration comes at the cost of overconfidence, as reflected in the higher Log Loss values. In contrast, Log Loss consistently favors the Logistic Regression model, which makes more accurate and confident predictions. This suggests that Logistic Regression is better suited for the problem, especially in high-stakes predictions where minimizing confident incorrect predictions is critical. Based on the results, the Logistic Regression model demonstrates superior performance and should be used for predicting player tournament outcomes.

6. APPLICATIONS OF THE MODEL

The predictive model can be applied in multiple areas of sports analytics, with a particular focus on forecasting player performance in PGA tournaments. For coaches and sports analysts, the model provides valuable insights into a player's likelihood of finishing in specific positions, such as making the cut or finishing in the top 10, allowing for more informed strategic decisions during tournaments. Additionally, it can be utilized in the world of sports betting, where accurate performance predictions can enhance decision-making and betting strategies. Fantasy sports platforms could also benefit from the model, offering users better predictions for player outcomes, improving team selection and roster management. Furthermore, the model can be leveraged by sponsors and agents to assess player performance trends over time, helping them make decisions about sponsorship deals or player development opportunities based on objective data-driven predictions.

7. NEXT STEPS

The model presents promising results in using prior player performance to predict future tournament outcomes and overall trends in play. However, there are several ways the model can be improved in future iterations. Incorporating more detailed and expansive data, such as round-by-round performance metrics, will likely enhance predictive accuracy. By using this finer-grained data, the model could better capture short-term variations in player form, leading to even more accurate predictions.

The current fixed effects approach, while effective for isolating player and tournament effects, assumes that player skill and tournament conditions are static over time. This limits the model's ability to capture dynamic trends like changes in player form, fatigue, or tournament-specific variations. Additionally, the assumption of additive and linear relationships between players and tournaments overlooks potential interactions, such as how specific players might perform better or worse in certain tournaments or on particular courses. Furthermore, the model does not account for external factors like weather, travel fatigue, or course-specific challenges that may influence performance.

To improve the model, future work should explore the introduction of time-varying effects to account for fluctuations in player performance, as well as course-specific effects to control for consistent conditions. Incorporating interaction terms would help capture non-linear relationships between players and tournaments. Mixed effects models could offer more flexibility in modeling random variability across players and tournaments.